

Software Process and Models

What is SW Process?

- When building a Product it's important to go through a series of predictable steps, road map, that help you create a timely, and high-quality result.
- The ROAD MAP that you follow is the SW process.
- A process is a collection of activities, actions, and tasks that are performed when some work product is to be created.
- Process depends on the SW you are building
 - Web design vs. telephone switch!!

Software Process

- Process defines who is doing:
 - – What?,
 - – When and how?
 - – To reach a goal
- Process defines tasks and activities within a schedule
- Results
 - – Programs
 - – Documents
 - – And data
- Software process is a roadmap for high quality software

A Process Framework

Software Process

Process framework

Framework activity 1

Framework activity n

Umbrella Activities

Process framework

Framework activities

work tasks

work products

milestones & deliverables

QA checkpoints

Umbrella Activities

A Process Framework

Process framework

Framework activities

work tasks

work products

milestones & deliverables

QA checkpoints

Umbrella Activities

Process framework

Modeling activity

Software Engineering action: **Analysis**

work tasks: requirements gathering,
elaboration, negotiation,
specification, validation

work products: analysis model and/or
requirements specification

milestones & deliverables

QA checkpoints

Software Engineering action: **Design**

work tasks: data design, architectural,
interface design,
component design

work products: design model
and/or design specification

Umbrella Activities

Framework Activities

- Communication
- Planning
- Modeling
 - Analysis of requirements
 - Design
- Construction
 - Code generation
 - Testing
- Deployment
 - Delivery
 - Feedback

■ Activity: Communication

■ Task set for a relatively simple project

- Make a list of stakeholders for project
- Invite all stakeholders for an informal meeting
- Ask each stakeholder to make a list of required features and functions
- Discuss requirements and build a final list
- Prioritize requirements
- Note areas of uncertainty

Activity: Communication , Action: requirement gathering

■ Task set for a larger project

- Make a list of stakeholders for project
- Interview each stakeholder separately
- Build a preliminary list of required features and functions
- Schedule a series of facilitated requirement gathering meetings
- Conduct meetings
- Produce informal user scenarios as part of each meeting
- Refine user scenarios based on stakeholder feedback
- Build a revised list of stakeholder requirements
- Use quality function deployment techniques to prioritize requirements
- Package requirements so that they can be delivered incrementally
- Note constraints and restrictions that will be placed on the system
- Discuss methods for validating the system

Typical Umbrella Activities

- Software project tracking and control
- Risk management
- Software quality assurance
- Formal technical reviews
- Measurement
- Software configuration management
- Reusability management
- Work product preparation and production

- Software project tracking and control
 - To maintain schedule ...
- Risk management – assess risk that may affect the outcome of project.
- Software quality assurance
- Formal technical reviews
 - Uncover and remove errors before they propagate to the next action
- Measurement
 - Defines and collect process, project and product measures
- Software configuration management
 - Manages the effects of change
- Reusability management –define criteria for work product reuse.
- Work product preparation and production
 - Create models, documents, logs, forms, lists ...

Generic Framework activities

■ Communication

- Get to know your Customer and their processes
- Identify stakeholders
- Requirement elicitation

Generic Framework activities

- Planning
 - Plan the work
 - Identify resources
 - Identify tasks
 - Set the schedule

Generic Framework activities

- Modeling
 - Analysis of requirements
 - Design
 - Blue print for customer and developers communications

Generic Framework activities

■ Construction

- Code generation
- Testing

■ Deployment

- Software is delivered to the customer.
- Customer evaluates and provides feedback.

Prescriptive (Traditional) Software Process Models

Prescriptive Models

- Prescriptive process models advocate an orderly approach to software engineering
 - Organize framework activities in a certain order
- Originally proposed to bring order to the chaos of software development
- They brought order to software engineering work and provide reasonable guidance to software teams
- Adopt following activities
 - Communication
 - Planning
 - Modeling
 - Construction
 - Deployment
- Differ in
 - Emphasis of these activities
 - Manners in which the framework is invoked and its relationship with other activities

SW Process Models

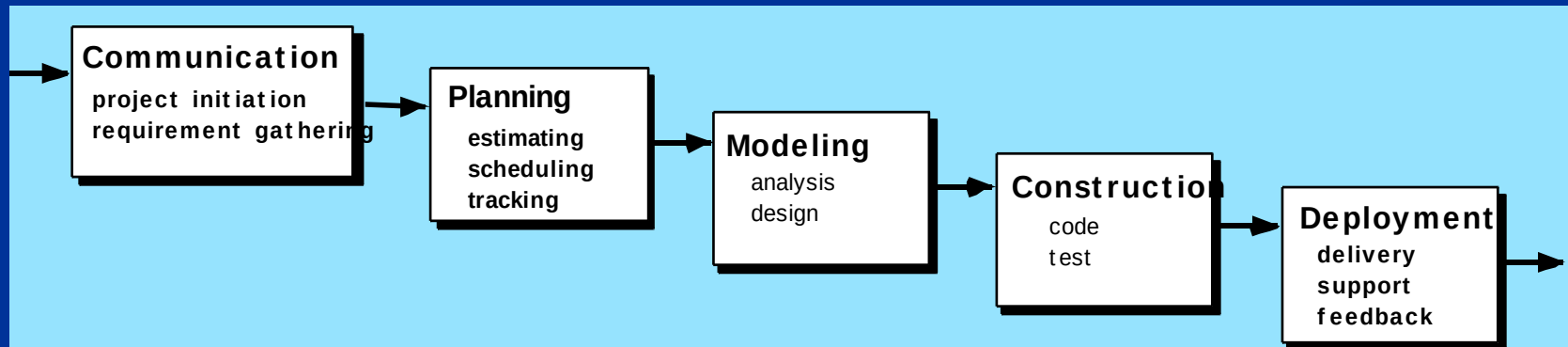
- Guides the SW team through a set of framework activities (process flow)
 - Linear
 - Incremental
 - evolutionary
- It is a set of **activities** required to
 - Define, design, implement, test and maintain a software product.
 - A SW process model is chosen based on the nature of the project.
 - Each process model prescribes a workflow for SE activities

SW Process Model Phases

- All models have phases and each phase has 3 components:
 - Set of **activities**, this is what you do.
 - Set of **deliverables**, this is what you produce.
 - **Quality control measures**, this is what you use to evaluate the deliverables.
- The activities defines the process Framework, the generic set encompasses:
 - Communication, planning, modeling, construction, and deployment

The Waterfall Model

This Model suggests a systematic, sequential approach to SW development that begins at the system level and progresses through analysis, design, code and testing.



Waterfall Model

Advantages

- Easy
- Structured
- Provide a template into which methods for analysis, design, code, testing and maintenance can be placed.

Disadvantages

- Sequential, does not reflect reality
- Does not allow for feedback
- Does not produce a prototype
- User must wait until the end to see the final program.

Limitations of the waterfall model

- The model implies that you should attempt to complete a given stage before moving on to the next stage
 - Does not account for the fact that requirements constantly change.
 - It also means that customers can not use anything until the entire system is complete.
- The model makes no allowances for prototyping.
- Assumes understanding of problem and full requirements early on
- It implies that you can get the requirements right by simply writing them down and reviewing them.
- Inflexible partitioning of the project into distinct

Critique of Waterfall Model **continued**

- Follows systematic approach to development
- The model implies that once the product is finished, everything else is maintenance.
- Assumes patience from customer
- Surprises at the end are very expensive
- Some teams sit ideal for other teams to finish
- Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.

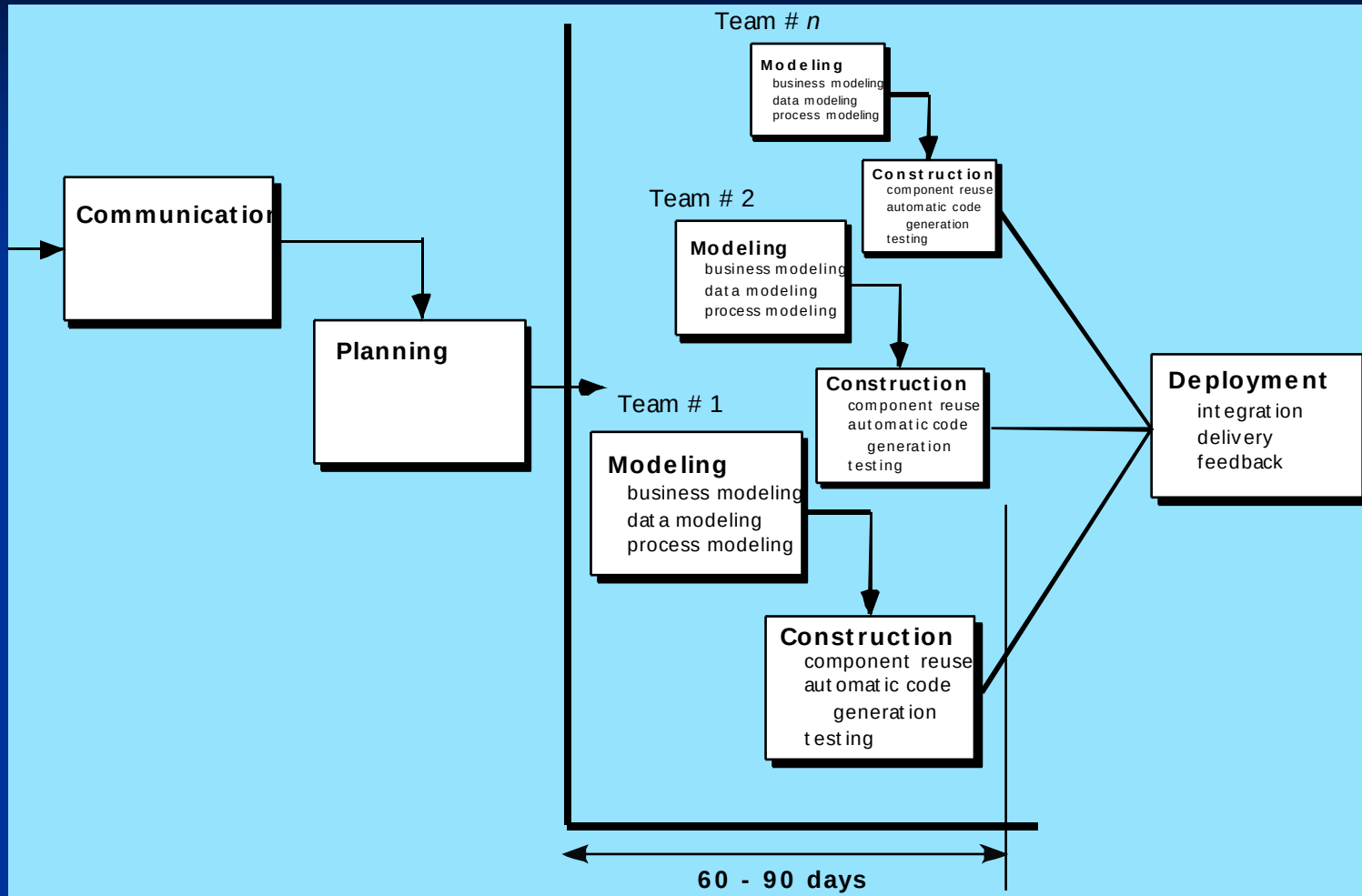
When to use the Waterfall Model

- Simple Projects
- Limited amount of time
- Requirements are well understood
- We can use it for our Class Project.

2. The Rapid Application Development RAD Model

- Builds on the Incremental model with emphases on short development cycle.
- In other words high speed waterfall model
- Components are build using this model as a fully functional units in a relatively short time
- It assumes that the system can be modularized
- RAD will fail if you don't have strong and skillful teams
- High performance might be an issue

The RAD Model



The Phases of RAD Model

■ Business modeling.

- The information flow among business functions is modeled in a way that answers the following questions:
 - What information drives the business process?
 - What information is generated?
 - Who generates it?
 - Where does the information go?
 - Who processes it?

The Phases of RAD Model

■ Data modeling.

- The information flow defined as part of the business modeling phase is refined into a set of data objects that are needed to support the business.
- The characteristics (called *attributes*) of each object are identified and the relationships between these objects defined.

The Phases of RAD Model

- **Process modeling.**
 - The data objects defined in the data modeling phase are transformed to achieve the information flow necessary to implement a business function.
 - Processing descriptions are created for adding, modifying, deleting, or retrieving a data object.

Drawbacks of RAD Model

- For large but scalable projects, RAD requires sufficient human resources to create the right number of RAD teams.
- RAD requires developers and customers who are committed to the rapid-fire activities necessary to get a system complete in a much abbreviated time frame. If commitment is lacking from either constituency, RAD projects will fail.

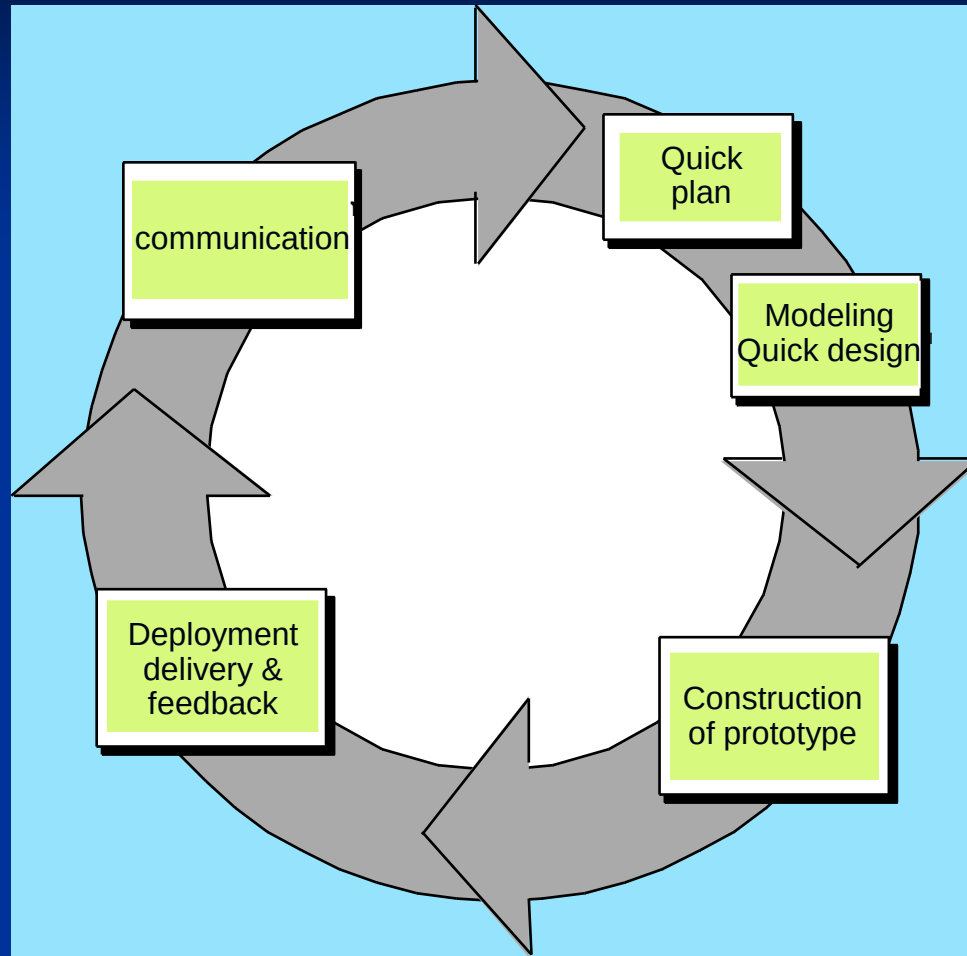
Drawbacks of RAD Model

- Not all types of applications are appropriate for RAD. If a system cannot be properly modularized, building the components necessary for RAD will be problematic. If high performance is an issue and performance is to be achieved through tuning the interfaces to system components, the RAD approach may not work.
- RAD is not appropriate when technical risks are high. This occurs when a new application makes heavy use of new technology or when the new software requires a high degree of interoperability with existing computer programs.

Prototyping Model

- Start with what is known about requirements.
- Do a quick design.
- Build the prototype by focusing on what will be seen by the user.
- Use the prototype to show the user and help refining requirements.

Evolutionary Models: Prototyping



When to Use Prototype Model

- When the customer define general objectives for the SW but does NOT identify details about INPUT, OUTPUT, or processing requirements.
- The developer is unsure of the efficiency of an algorithm, human machine interaction, etc.

Prototype Model

Advantages

- Prototype is served as the machinery for identifying requirements.
- Is developed very quick.

Disadvantages

- Customer might think that the prototype is the final product and forget lack of quality i.e PERFORMANCE, RELIABILITY.
- The developer often makes implementation compromises in order to get a prototype working quickly.

Evolutionary Process Models

- Core requirements are well understood but additional requirements are evolving and changing fast
- Time-to-Market
- Iterative – software gets more complex with each iteration
 1. Incremental
 2. Spiral
 3. WIN-WIN Spiral Model
 4. Concurrent Development Model

Evolutionary SW Process Models

■ Advantages

- Do not require **full** knowledge of the requirements
- **Iterative**
- Divide project into **builds**
- Allows feedback, show user something sooner
- Develop more complex systems

Incremental Model

- Goal to provide quick basic functionality to the users
- Process is not linear
- Requirements are well defined
- Software is completed in an increments fashion

1. Incremental Model

- It **combines** characteristics of the waterfall model and the iterative nature of the prototyping model.
- 1st build is usually the **CORE** product
- Each increment “deliverable” may **add a new** functionality.
- This is repeated until the product is complete
- Delivers software in small but usable pieces, each piece builds on pieces already delivered

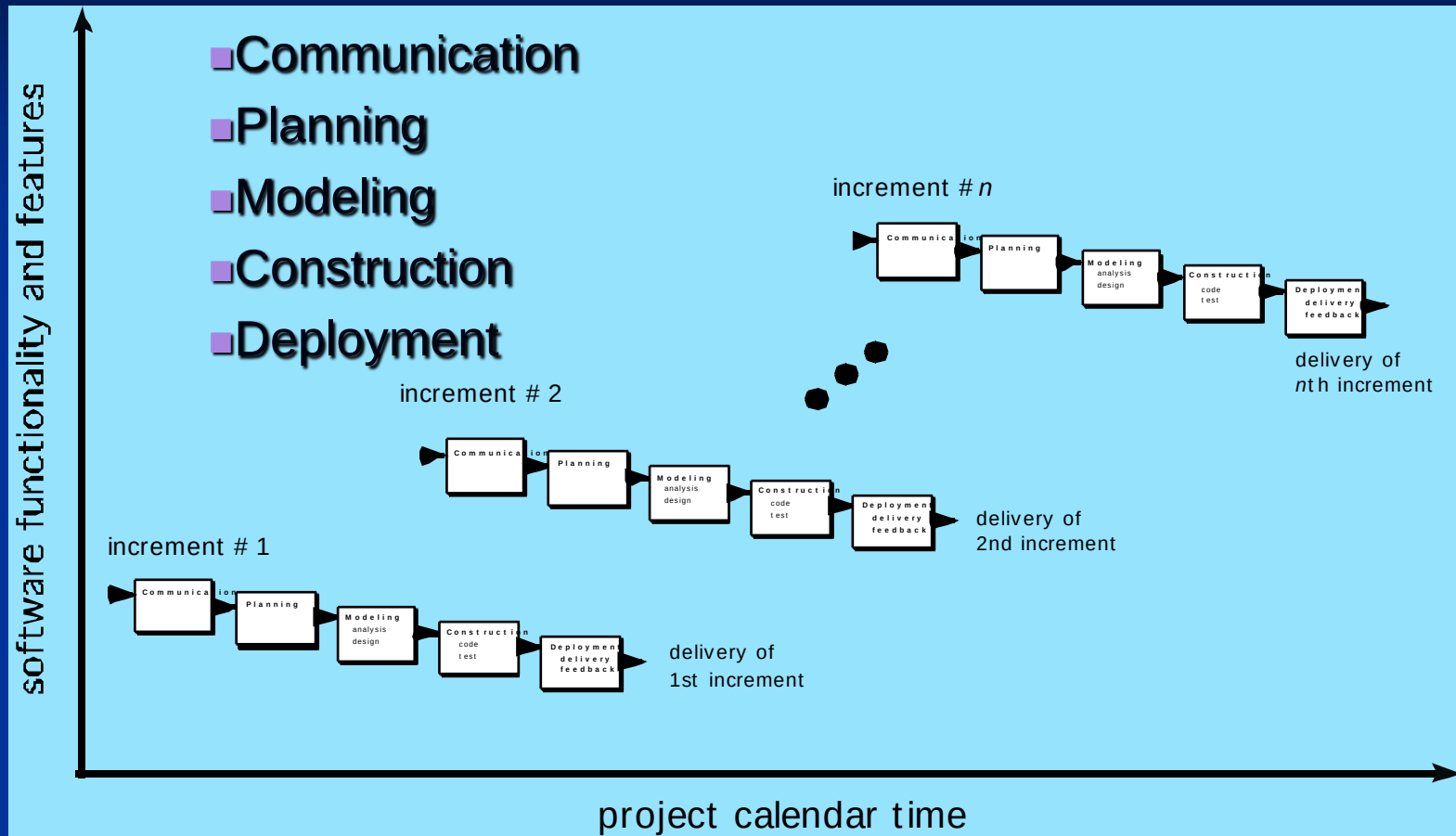
The Incremental Model

- Rather than deliver the system as a single delivery, the development and delivery is broken down into increments with each increment delivering part of the required functionality.
- First Increment is often core product
 - Includes basic requirement
 - Many supplementary features (known & unknown) remain undelivered
- First Increment is used or evaluated
- A plan of next increment is prepared
 - Modifications of the first increment
 - Additional features of the first increment
- It is particularly useful when enough staffing is not available for the whole project
- Increment can be planned to manage technical risks

The Incremental Model

- User requirements are prioritised and the highest priority requirements are included in early increments.
- Once the development of an increment is started, the requirements are frozen though requirements for later increments can continue to evolve.
- Customer value can be delivered with each increment so system functionality is available earlier.
- Early increments act as a prototype to help elicit requirements for later increments.
- Lower risk of overall project failure.
- The highest priority system services tend to receive the most testing.

The Incremental Model



Example

- For example, word-processing software developed using the incremental paradigm might deliver basic file management, editing, and document production functions in the first increment;
- more sophisticated editing and document production capabilities in the second increment;
- spelling and grammar checking in the third increment;
- and advanced page layout capability in the

When to Use the Incremental Model

- When staffing is not available by deadline.
- When the software can be broken into increments and each increment represent a solution

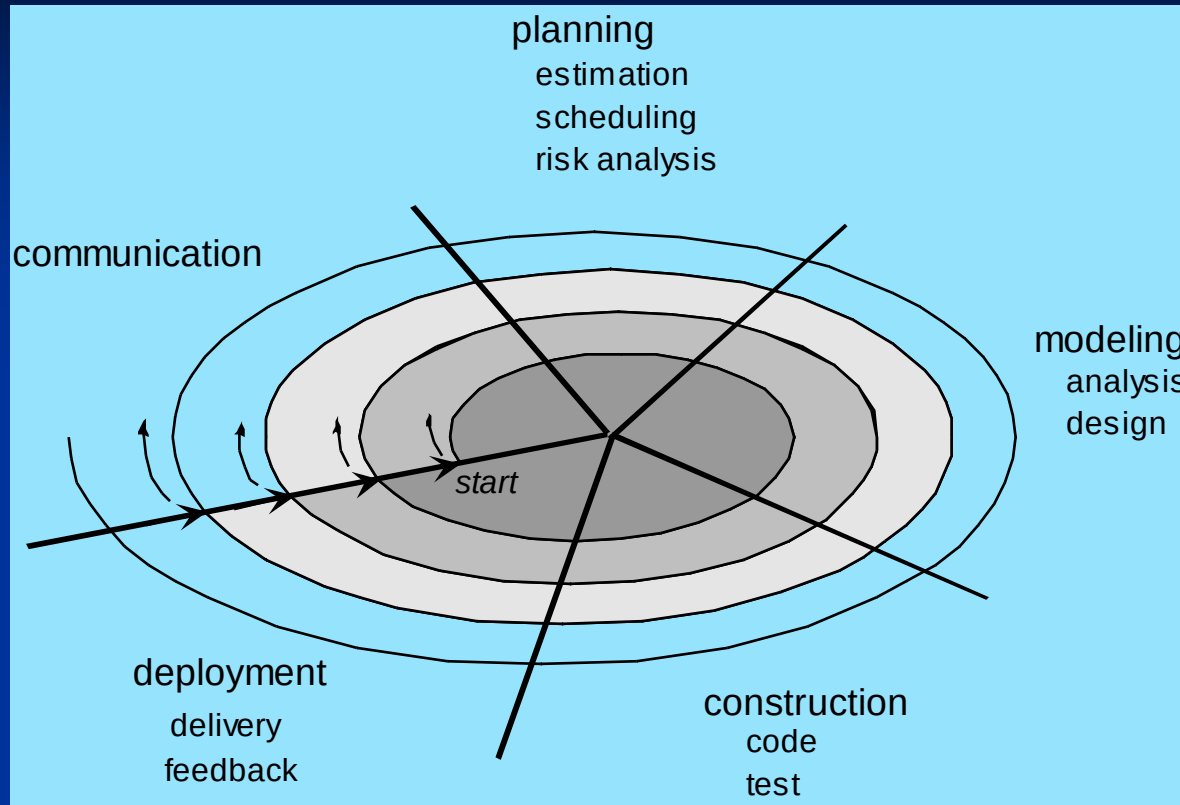
2. Spiral Model

- Iterative (like Prototype) and controlled (like waterfall) model.
- Software is developed using evolutionary releases
- Software complexity increase with each release

Spiral Model

- Consist of 6 task regions.
 - Customer communication - the goal is to establish good communication between customer and developer.
 - Planning - produce/adjust project plan.
 - Risk analysis - assess management and technical risks.
 - Engineering - build one or more representations of the application.
 - Construction and release - - to construct, test, install and support the application.
 - Customer evaluation – get customer feedback.

Evolutionary Models: The Spiral



Product maintenance projects
Product enhancement projects
New product development projects
Inner Spiral-- Concept development projects⁴⁶

When to Use the Spiral Model

- Very large projects.
- When technical skills must be evaluated at each step.

Questions..

- Compare waterfall and spiral model. [M-2010]
- Compare waterfall and spiral model for software development [M-2011]
- Explain RAD and spiral model for Software development [D-2011]
- Explain in detail RAD software life cycle model [M-2012]
- Discuss waterfall and spiral model of software development with merits and demerits.[D-2012]
- More questions...refer class notes

- END OF TOPIC

- NEXT TOPIC – AGILE PROCESS